Take Home Activity 04

Q1.

Car.java > Car

```java
public class Car {
    int year;
    String brand;
    double price;

    public Car() {
        year = 2000;
        brand = "Unknown";
        price = 0.00;
    }

    public void displayInfo() {
        System.out.println("Year: " + year);
        System.out.println("Brand: " + brand);
        System.out.println("Price: " + price);
    }

    Run | Debug
    public static void main(String[] args) {
        Car car1 = new Car();

        car1.displayInfo();
    }
}
```

```
D:\NSBM\NSBM\Year 1\S
Year: 2000
Brand: Unknown
Price: 0.0
```

Q2.

Car.java > Car

```java
public class Car {
    int year;
    String brand;
    double price;

    public Car() {
        year = 2000;
        brand = "Unknown";
        price = 0.00;
    }

    public void displayInfo() {
        System.out.println("Year: " + year);
        System.out.println("Brand: " + brand);
        System.out.println("Price: " + price);
    }

    Run | Debug
    public static void main(String[] args) {
        Car car1 = new Car();

        car1.displayInfo();
    }
}
```

```
D:\NSBM\NSBM\Year 1\Semester 2\Java\Assessment\Assessment 4>jav
My anme is john, StudentId is: 1, age is 25, GPA is: 13.0
My anme is due, StudentId is: 2, age is 45, GPA is: 12.0
GPA is update to: 0.56
```

Q3.

```java
Product.java > Product
public class Product{
    int productId;
    String productName;
    double unitPrice;

    public Product(){
        productId = 0;
        productName = "Unknown Product";
        unitPrice = 0.0;
    }

    public Product(int productId,String productName,double unitPrice){
        this.productId = productId;
        this.productName = productName;
        this.unitPrice = unitPrice;
    }

    public void applyDiscount(double presentage){
        unitPrice = unitPrice - (unitPrice*presentage/100);

        System.out.println("Updated price after: "+presentage+"% discount: "+unitPrice);
    }

    public String getProductLabel() {
        return productId + " - " + productName + " - Rs. " + unitPrice;
    }

    Run | Debug
    public static void main(String[] args) {
        Product product1 = new Product();

        Product product2 = new Product(productId: 101, productName: "Laptop", unitPrice: 150000.00);

        product2.applyDiscount(presentage: 15);

        System.out.println(product1.getProductLabel());
        System.out.println(product2.getProductLabel());
    }
}
```

```
D:\NSBM\NSBM\Year 1\Semester 2\Java\Assessment\Assessmer
Updated price after: 15.0% discount: 127500.0
0 - Unknown Product - Rs. 0.0
101 - Laptop - Rs. 127500.0
```

Q4.

```java
Product.java > Product
public class Product{
    int productId;
    String productName;
    double unitPrice;

    public Product(){
        productId = 0;
        productName = "Unknown Product";
        unitPrice = 0.0;
    }

    public Product(int productId,String productName,double unitPrice){
        this.productId = productId;
        this.productName = productName;
        this.unitPrice = unitPrice;
    }

    public void applyDiscount(double presentage){
        unitPrice = unitPrice - (unitPrice*presentage/100);

        System.out.println("Updated price after: "+presentage+"% discount: "+unitPrice);
    }

    public String getProductLabel() {
        return productId + " - " + productName + " - Rs. " + unitPrice;
    }

    Run | Debug
    public static void main(String[] args) {
        Product product1 = new Product();

        Product product2 = new Product(productId: 101, productName: "Laptop", unitPrice: 150000.00);

        product2.applyDiscount(presentage: 15);

        System.out.println(product1.getProductLabel());
        System.out.println(product2.getProductLabel());
    }
}
```

```
D:\NSBM\NSBM\Year 1\Semester 2\Java\Assessment\Assessment 4>
Employee ID: 101, Name: gimahana, Department: IT
Employee ID: 102, Name: noone, Department: HR
Net Salary of gimahana: Rs. 96000.0
Net Salary of noone: Rs. 84000.0
gimahana has been promoted to Management.
```

Q5.

BankAccount.java > BankAccount > main(String[])

```java
class BankAccount {
    int accountNo;
    String holderName;
    double balance;

    BankAccount() {
        accountNo = 1001;
        holderName = "Default User";
        balance = 5000.0;
    }

    BankAccount(int accountNo, String holderName, double balance) {
        this.accountNo = accountNo;
        this.holderName = holderName;
        this.balance = balance;
    }

    void deposit(double amount) {
        balance += amount;
        System.out.println("Updated Balance: " + balance);
    }

    String withdraw(double amount) {
        if (balance >= amount) {
            balance -= amount;
            return "Withdrawal successful. Remaining balance: " + balance;
        } else {
            return "Insufficient funds.";
        }
    }

    double getBalance() {
        return balance;
    }

    String getAccountInfo(String branch) {
        return "Account No: " + accountNo +
               ", Holder Name: " + holderName +
               ", Branch: " + branch;
    }

```

```java
    Run | Debug
    public static void main(String[] args) {
        BankAccount acc1 = new BankAccount();

        BankAccount acc2 = new BankAccount(accountNo: 2002, holderName: "John Silva", balance: 10000.0);

        System.out.println(x: "---- Account 1 ----");
        acc1.deposit(amount: 1500);
        System.out.println(acc1.withdraw(amount: 2000));
        System.out.println("Current Balance: " + acc1.getBalance());
        System.out.println(acc1.getAccountInfo(branch: "Colombo Branch"));

        System.out.println(x: "\n---- Account 2 ----");
        acc2.deposit(amount: 2500);
        System.out.println(acc2.withdraw(amount: 5000));
        System.out.println("Current Balance: " + acc2.getBalance());
        System.out.println(acc2.getAccountInfo(branch: "Kandy Branch"));
    }
}
```

```
D:\NSBM\NSBM\Year 1\Semester 2\Java\Assessment\Assessment 4>java BankAccou
---- Account 1 ----
Updated Balance: 6500.0
Withdrawal successful. Remaining balance: 4500.0
Current Balance: 4500.0
Account No: 1001, Holder Name: Default User, Branch: Colombo Branch

---- Account 2 ----
Updated Balance: 12500.0
Withdrawal successful. Remaining balance: 7500.0
Current Balance: 7500.0
Account No: 2002, Holder Name: John Silva, Branch: Kandy Branch
```